California Traffic Collision Data Analysis

PySpark ETL and evidence report | 3 November 2026

Executive summary

The saved notebook analyses 935,791 sampled California collision records, with separate party and victim files. Property damage only accounts for 60.96% of recorded collisions and severe injury for 2.37%. Los Angeles has the highest county count (284,100). Friday, October and 17:00 have the highest recorded counts in their respective comparisons. These are frequencies in the supplied sample, not exposure-adjusted measures of road risk.

The strongest segmented finding is that 4.35% of collisions recorded in darkness without street lights were classified as severe injury, compared with 1.92% in daylight. This is an association that warrants a location-specific safety audit; lighting alone cannot be identified as the cause. Processed collision, party and victim tables were exported to Amazon S3 as Parquet, and the notebook verified 600 uploaded files.

1 Data and analytical scope

The starter notebook identifies the source as SWITRS-derived California traffic data. The analysis actually reads four sampled CSV files from Google Drive rather than the full SQLite database described in the introductory text. The collision file is the basis for collision-level findings; the party and victim files are analysed separately because the sample linkage has not been validated.

Input table	Rows	Role in report
Collisions	935,791	Severity, geography, weather, lighting and time
Parties	1,866,917	Age cleaning and outlier assessment
Victims	963,933	Victim age distribution
Case IDs	942,433	Loaded for preparation; not joined into the analysis

The collision sample spans 2001 to 2021. PySpark 4.0.5 and pandas 2.2.3 were used in the saved Colab run. The report quotes saved cell outputs; it does not imply that the sample represents every California collision.

2 Preparation and data quality

2.1 Cleaning decisions

All CSV fields were initially read as strings. The notebook standardised empty strings and text placeholders as missing values, converted dates, counts, coordinates and ages to appropriate types, and retained case IDs as text. Missing values in selected categorical fields were labelled “Unknown”; missing numerical values, dates and coordinates remained null rather than being treated as zero. No records were removed solely for missingness.

No collision column reached the notebook’s 95% missingness threshold, so all 65 collision columns were retained. Latitude and longitude each had 669,049 missing values (71.50%). This substantially limits any map or hotspot analysis. After age validation, 931,758 victim records had a valid age and 32,175 had a missing or invalid age.

2.2 Duplicate and outlier checks

Exact-duplicate checks removed no rows from the collision, party or victim samples. Twenty-seven collision case IDs appeared more than once, so the notebook retained those records rather than assuming that every repeated ID was an identical event. This is an unresolved linkage and data-quality issue.

Approximate IQR checks flagged 4,548 party ages above the upper bound of 87.5 years, 4,568 victim ages above 85.5 years and 33,313 collision rows with more than 2.5 injured victims. Those events can be plausible, so the notebook retained them. Invalid ages below 0 or above 120 were set to null. The party-count IQR was zero, so the rule was not useful for that field.

3 Exploratory findings

3.1 Severity, weather and victim age

Of the collision rows, 570,480 were property damage only (60.96%), 226,492 pain injury (24.20%), 116,681 other injury (12.47%) and 22,138 severe injury (2.37%). This distribution provides the baseline for the segmented comparisons below.

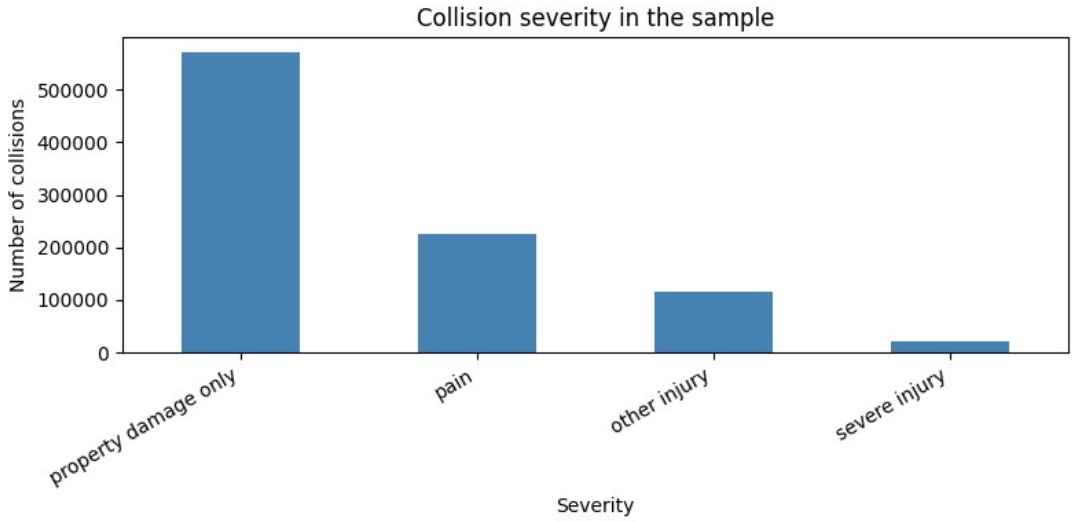


Figure 1. Distribution of recorded collision severity in the sampled collision table.

Clear weather appears in 769,927 records (82.28%), cloudy weather in 122,155 (13.05%) and rain in 32,341 (3.46%). Higher counts in clear weather cannot be read as higher risk without driving exposure by weather condition. Valid victim ages are concentrated among younger adults: ages 18, 19 and 20 have the three highest individual counts in the separate victim sample.

3.2 Severity by victims, weather and lighting

Where the injured-victim count is known, the mean recorded injured victims per collision rises from 1.37 for pain injury to 1.51 for other injury and 1.62 for severe injury. The median is one in all three injury categories. This is a collision-level comparison and does not require joining the separately sampled victim file.

Within recorded weather groups, the severe-injury share is 2.42% in clear conditions, 1.92% in rain and 3.59% in fog. Fog has 3,901 collision records; wind also shows 3.59% but has only 390 records, so its percentage is especially unstable. The analysis describes association, not a weather effect adjusted for travel volume, road type or speed.

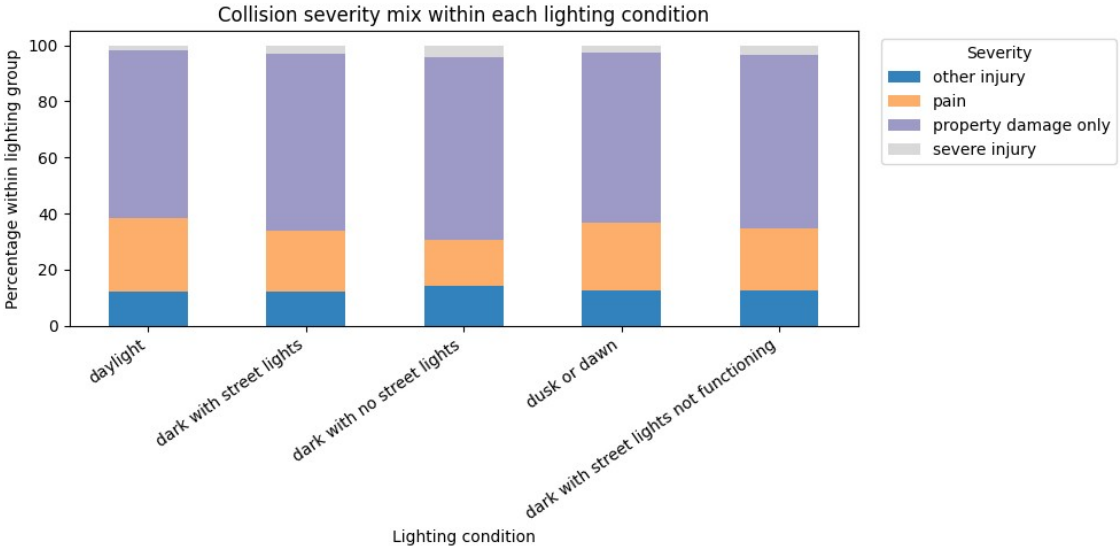


Figure 2. Severity mix within each lighting group. Percentages use each lighting group as the denominator.

The severe-injury share is 4.35% without street lights, 2.97% with working street lights and 1.92% in daylight. The comparison supports prioritising detailed audits of dark locations, while recognising differences in traffic mix, speed, road design and travel exposure.

3.3 Geography and recorded time patterns

Los Angeles accounts for 284,100 recorded collisions (30.36% of records with a known county), followed by Orange (72,042), San Bernardino (56,737), San Diego (53,105) and Riverside (48,686). County totals reflect both the sample and the amount of travel; they do not establish which county has the greatest per-driver risk.

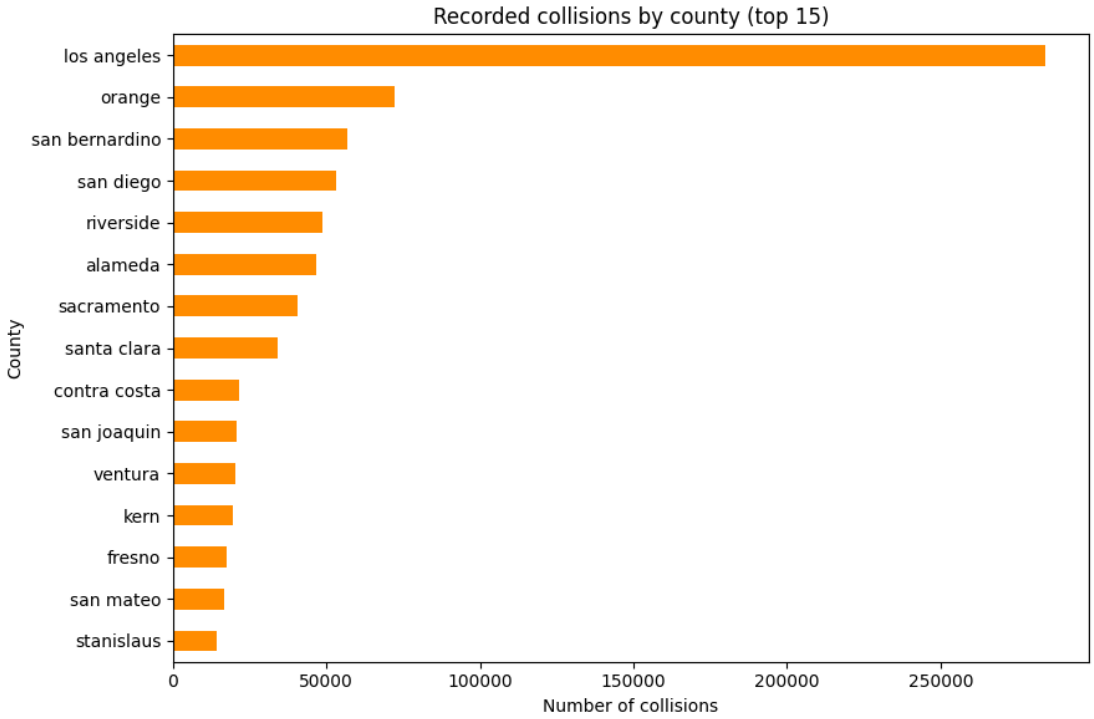


Figure 3. The 15 counties with the highest collision counts in the sampled data.

Only 266,742 of 935,791 collision records have valid coordinates. The notebook plots a reproducible 10% sample of these locations (26,868 points), with extreme coordinate values outside the display limits. The spatial picture is therefore incomplete and should not be used alone to select precise intervention sites.

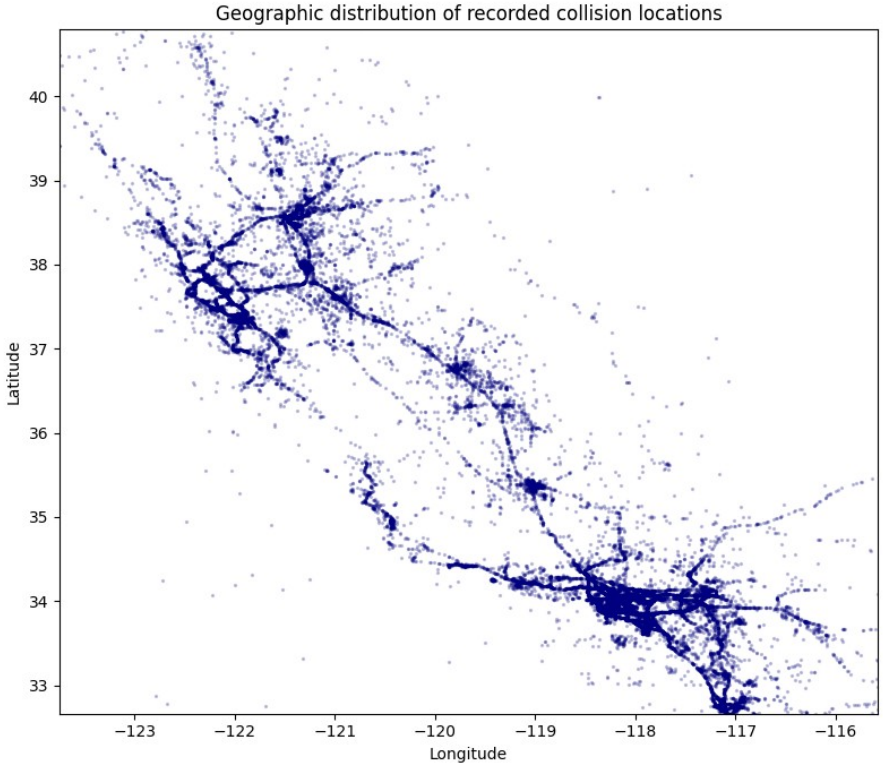


Figure 4. Geographic scatter plot of a 10% sample of valid geocoded collision records.

Friday has the largest weekday count (154,544), October the largest calendar-month count (83,274) and 17:00 the largest hourly count (73,255). Annual counts range from 53,708 in 2002 to 13,198 in 2021, but uneven coverage and a potentially incomplete final year prevent a simple statewide trend claim.

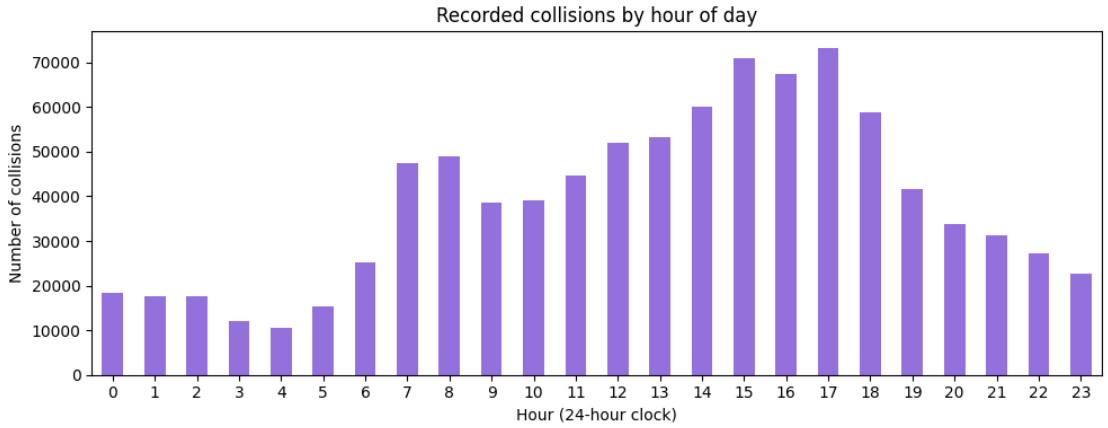


Figure 5. Recorded collisions by hour. Counts have not been adjusted for hourly traffic exposure.

4 ETL query results

The seven Spark SQL questions in the notebook return the following results. “Highest” and “most dangerous” refer to observed frequency in this sample, not causal contribution or individual risk.

Question	Saved result
Top five counties	Los Angeles 284,100; Orange 72,042; San Bernardino 56,737; San Diego 53,105; Riverside 48,686
Highest month	October: 83,274
Most common weather	Clear: 769,927
Recorded fatal collisions	0 of 935,791 (0.0000%); killed-victim count missing in 200 rows
Highest-count hour	17:00: 73,255
Top road surfaces	Dry 845,637; wet 76,833; snowy 4,122; slippery 1,048; “H” 10
Top lighting groups	Daylight 625,574; dark with street lights 195,867; dark without street lights 74,634

The “H” road-surface label should be checked against the source codebook before interpretation. The zero fatal-collision result describes this sample and the query definition only. It is not evidence that no fatal collisions occurred statewide, especially because the sample contains 200 unknown killed-victim counts.

5 Implications and recommendations

Location audits. Start with high-count counties and use validated local collision and travel data to identify specific junctions or corridors. Prioritise sites with repeated severe injury, then review speed, crossings, junction geometry and signal operation.

Darkness and severity. Review lighting, speed limits and road design at locations with severe injuries in darkness, particularly where street lights are absent or not working. Compare injury outcomes before and after any change and use matched sites where feasible.

Peak periods. Use Friday and 17:00 counts to choose periods for closer traffic-management review. Check vehicle and pedestrian volumes before changing signal timing or enforcement schedules.

Vulnerable road users. The notebook contains pedestrian and cyclist collision fields, but it does not report separate trends for those groups. Produce dedicated counts, severity rates and location maps before choosing crossing upgrades or cycle protection measures.

Predictive work. After validating identifiers and geocodes, train a hotspot model on earlier years and evaluate it on later years and held-out locations. Include traffic exposure and test whether it improves targeting over a simple historical-count baseline.

6 Limits and conclusion

The main limitations are the unvalidated relationship between separately sampled tables, 27 repeated case IDs, 71.50% missing coordinates, unknown traffic exposure, potentially uneven annual coverage, and the absence of recorded fatalities in the collision sample. Categories are not causal explanations: clear weather and daylight have high counts partly because driving is common in those conditions. The geographic scatter plot is illustrative, not a complete hotspot map.

The project establishes a reproducible PySpark workflow from CSV loading and cleaning through descriptive analysis, Spark SQL queries and Parquet export. Its strongest actionable result is the higher observed severe-injury share in unlit darkness; the next step is to validate that signal with location-level and exposure data before committing to infrastructure changes. The S3 export is complete at the level of uploaded object listing, with 600 Parquet files verified across the three processed tables.

7 AWS S3 upload evidence

The processed collision, party and victim tables were written as Parquet under the private bucket nachiket-crash-analysis-2004-2707 in the ap-southeast-2 region. The Colab upload cell reported successful uploads for all three tables and verified 600 Parquet object keys. The S3 console screenshot below independently shows the three table prefixes. The screenshot shows folders, while the file count comes from the notebook output; no read-back integrity test is claimed.

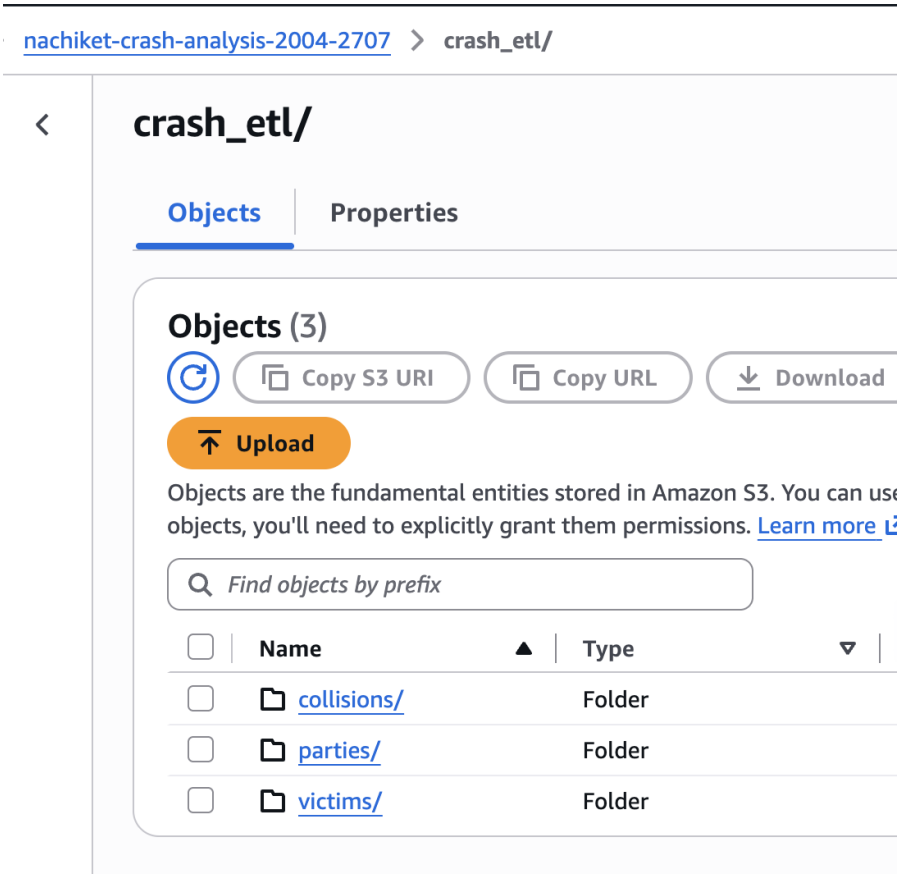


Figure 6. AWS S3 console showing the crash_etl prefix with collisions, parties and victims folders, 3 November 2026.